



Nearly tight approximation algorithm for (connected) Roman dominating set

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Abstract

A Roman dominating function of graph G is a function $r : V(G) \rightarrow \{0, 1, 2\}$ satisfying that every vertex v with $r(v) = 0$ is adjacent to at least one vertex u with $r(u) = 2$. The minimum Roman dominating set problem (MinRDS) is to compute a Roman dominating function r that minimizes the weight $\sum_{v \in V} r(v)$. The minimum connected Roman dominating set problem (MinCRDS) is to find a minimum weight Roman dominating function r_c such that the subgraph of G induced by $D_R = \{v \in V \mid r_c(v) = 1 \text{ or } r_c(v) = 2\}$ is connected. In this paper, we present a greedy algorithm for MinRDS with a guaranteed performance ratio at most $H(\delta_{\max} + 1)$, where $H(\cdot)$ is the Harmonic number and $\delta_{\max}$ is the maximum degree of the graph. For any $\varepsilon > 0$, we show that there exists a greedy algorithm for MinCRDS with approximation ratio at most $(1 + \varepsilon) \ln \delta_{\max} + O(1)$. The challenge for the analysis of the MinCRDS algorithm lies in the fact that the potential function is not only non-submodular but also non-monotone.

Keywords Roman dominating set · Connected Roman dominating set · Non-submodular optimization · Greedy algorithm · Approxiamtion ratio

1 Introduction

The concept of Roman domination comes from an article titled Defend the Roman Empire published by Stewart in 1999 [15]. A region is considered safe if some legion is stationed there. Otherwise it is considered insecure. An insecure region could be

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secured if it can be reached by a legion from a neighboring region. In the fourth century B.C., the supreme ruler issued a decree that a region was forbidden to send legion to an insecure area if it leaves its original region unprotected. Therefore, only the areas with two legions are allowed to send out legion. Since it is expensive to maintain a legion at a location, the Emperor would like to station as few legions as possible, while still defending the Roman Empire.

Cockayne et al. [3] formulated this problem on a graph $G = (V, E)$ as follows. A function $r : V(G) \rightarrow \{0, 1, 2\}$ is called a *Roman dominating function* if every vertex v with $r(v) = 0$ is adjacent to at least one vertex u with $r(u) = 2$. Let V_r^0, V_r^1, V_r^2 represent the sets of vertices with values 0, 1, 2 under the function r , respectively. Clearly, $|V_r^0| + |V_r^1| + |V_r^2| = |V|$. Denote by $r(V) = \sum_{v \in V} r(v)$ the weight of the Roman dominating function r . The *Roman domination number* $\gamma_R(G)$ is the minimum weight of a Roman dominating function of G . The *minimum Roman dominating set problem* (MinRDS) is to find a minimum weight Roman dominating function r for G . In some case, it is desirable that the legions can communicate smoothly, where communication is guaranteed by connectivity. Such a consideration leads to the *minimum connected Roman dominating set problem* (MinCRDS), the goal of which is to compute a minimum Roman dominating function r_c such that the subgraph of G induced by $V_{r_c}^1 \cup V_{r_c}^2$ is connected.

We only know two papers studying approximation algorithm for MinRDS in a general graph. Pagourtzis et al. [12] gave a $(2 + 2 \ln n)$ -approximation algorithm and Padamutham et al. [11] gave a $(2 + 2 \ln(\delta_{\max} + 1))$ -approximation algorithm, where $\delta_{\max}$ is the maximum degree of the graph. For unit disk graph, Shang et al. [14] presented a 5-approximation algorithm for MinRDS and a 7.5-approximation algorithm for MinCRDS, respectively. Note that ratio $(2 + 2 \ln(\delta_{\max} + 1))$ is asymptotically away from the possible lower bound of approximation ratio for MinRDS by a factor about 2 (see [11]), and as far as we know, there is no approximation algorithm for MinCRDS in a general graph. In this paper, we present an $H(\delta_{\max} + 1)$ -approximation algorithm for MinRDS and a $((1 + \varepsilon) \ln \delta_{\max} + O(1))$ -approximation algorithm for MinCRDS in a general graph, respectively.

1.1 Related works

Both MinRDS and MinCRDS are NP-hard in general graphs [5]. Recently, Padamutham et al. [11] had shown that even for star convex bipartite graphs and comb convex bipartite graphs, MinRDS cannot be approximated within factor $(1 - \varepsilon) \ln |V|$ for any $\varepsilon > 0$ unless $NP \subseteq DTIME(|V|^{O(\log \log |V|)})$. On the other hand, MinRDS can be solved in linear time on some special graphs, such as trees [12] and interval graphs [8].

The Roman domination number is extensively studied from graph theoretical point of view [1–4, 7, 16]. There are relatively less work on approximation algorithms for the minimum Roman domination problem. For MinRDS on a general graph, Pagourtzis et al. [12] proposed a $(2 + 2 \ln n)$ -approximation algorithm. This ratio was improved to $(2 + 2 \ln(\delta_{\max} + 1))$ by Padamutham et al. [11]. The method in [11] is as follows: first compute a dominating set D , using a $(1 + \ln(\delta_{\max} + 1))$ -approximation algorithm;

then assign value 2 to every vertex in D and value 0 to the other vertices, leading to an increase in the approximation ratio by a factor 2. For unit disk graphs, Shang et al. showed in [14] that both MinRDS and MinCRDS are still NP-hard, but admit 5-approximation and 7.5-approximation, respectively, by exploring relation between a maximal independent set and a minimal Roman dominating set. We have not seen any paper working on approximation algorithm for MinCRDS in general graphs.

For some recent algorithmic studies on variants of Roman domination problem, refer to [9, 10, 13, 17].

1.2 Our contribution

The contribution of this paper is summarized as follows.

- (i) We present a greedy algorithm for MinRDS in a general graph, which achieves approximation ratio $H(\delta_{\max} + 1)$, where $\delta_{\max} > 0$ is the maximum degree of the graph. It is well known that if the potential function f used in a greedy algorithm is monotone increasing and submodular, then an $H(\gamma)$ -approximation ratio can be achieved [6], where $\gamma = \max\{f(v) : v \in V\}$. In this paper we find a monotone increasing submodular potential function for MinRDS which results in approximation ratio $H(\delta_{\max} + 1)$. Note that $H(\delta_{\max} + 1) \leq 1 + \ln(\delta_{\max} + 1)$. Our result improves the previous ratio $(2 + 2 \ln(\delta_{\max} + 1))$ in [11] by a factor 2. In view of the inapproximability of this problem, our approximation ratio is best possible unless $P = NP$.
- (ii) We present a $((1 + \varepsilon) \ln \delta_{\max} + O(1))$ -approximation algorithm for MinCRDS, where $\varepsilon > 0$ is an arbitrary constant. The strategy is also greedy, with a delicately designed potential function. In each iteration, at most a constant number of vertices are added into current set, where “add” is not in the classic sense of union. In fact, we allow each vertex to be selected at most twice. A vertex which is selected $\ell \in \{0, 1, 2\}$ times has r_c -value ℓ . Note that a vertex might not get its value once for all: it can be selected at some time, getting value 1, and be selected another time, and thus having its value lifted from 1 to 2. The r_c -value assigned to a vertex may be viewed as its cost in the cost-effectiveness of the greedy choice. But different from a classic greedy algorithm, the cost is not predetermined, but is *dynamically changing* with the progress of the algorithm. This brings a big challenge to the analysis. Another challenge to the analysis is that the potential function we construct is neither monotone non-increasing nor monotone non-decreasing, and it is non-submodular. However, we managed to derive approximation ratio at most $(1 + \varepsilon) \ln \delta_{\max} + O(1)$. For this purpose, a crucial step is to construct a suitable decomposition of an optimal solution to bridge the value of an optimal solution and the computed solution. This is the first approximation algorithm for MinCRDS.

The rest of this paper is organized as follows. In Sect. 2, we present the algorithm for MinRDS and gives its theoretical analysis. Then we consider MinCRDS in Sect. 3. Section 4 concludes the paper and discusses some future research directions.

2 Greedy algorithm for MinRDS

2.1 Preliminaries

Without loss of generality, we assume all graphs considered in this paper are connected. Let $G = (V, E)$ be graph with vertex set $V = V(G)$ and edge set $E = E(G)$. Denote by $\delta_{\max}$ the maximum degree of G . The *open neighborhood* of a vertex $v \in V$ is the set $N_G(v) = \{u \in V \mid uv \in E(G)\}$ and the *closed neighborhood* of a vertex $v \in V$ is the set $N_G[v] = N_G(v) \cup \{v\}$. The *open neighborhood* of a set $C \subseteq V$ is the set $N_G(C) = \{\bigcup_{v \in C} N_G(v)\} \setminus C$. The *closed neighborhood* of a set $C \subseteq V$ is the set $N_G[C] = N_G(C) \cup C$. When the graph G is clear from the context, we shall omit the subscript G for simplicity of notation. For a vertex set $C \subseteq V \setminus \{v\}$, denote by $\deg_C(v)$ the number of neighbors of v in C . In particular, $\deg_G(v) = |N_G(v)|$ is the *degree* of v in G . Denote by $\delta_{\max} = \max\{\deg_G(v) : v \in V(G)\}$ the *maximum degree* of G . The subgraph of G induced by vertex set C , denoted as $G|_C$, is the graph with vertex set $V(G|_C) = C$ and edge set $E(G|_C) = \{uv : u, v \in C \text{ and } uv \in E(G)\}$. We say that a vertex set is connected if it induces a connected subgraph.

For a finite ground set U , we say that a set function $f : 2^U \rightarrow \mathbb{Z}$ is *submodular* if

$$f(C) + f(D) \geq f(C \cup D) + f(C \cap D) \text{ holds for any subsets } C, D \subseteq U.$$

We say that f is *monotone nondecreasing* if

$$f(C) \leq f(D) \text{ holds for any subsets } C \subseteq D.$$

A set function f is *normalized* if $f(\emptyset) = 0$. A *polymatroid* is a normalized monotone nondecreasing submodular set function.

In a *minimum submodular cover problem* (MinSMC), given a ground set U , a cost function $c : U \mapsto \mathbb{R}^+$ and a polymatroid $f : 2^U \mapsto \mathbb{R}^+$, the goal is to find a minimum cost element set $C \subseteq U$ satisfying

$$c(C) = \arg \min\{c(D) : D \subseteq U, f(D) = f(U)\}.$$

For two subsets $A, C \subseteq U$, denote by

$$\Delta_A f(C) = f(C \cup A) - f(C)$$

the *marginal profit* of A over C . In particular, when A consists of only one element x , we use $\Delta_x f(C)$ to abbreviate $\Delta_{\{x\}} f(C)$.

Wolsey [18] analyzed the following greedy algorithm for MinSMC: starting from $C = \emptyset$, iteratively add an element $x \in U$ into C which maximizes $\Delta_x f(C)/c(x)$, until $\Delta_x f(C) = 0$ for any $x \in U$ (the termination condition is equivalent with $f(C) = f(U)$, which is guaranteed by the fact that f is a polymatroid). Its approximation ratio is as follows.

Theorem 2.1 [18] *The greedy algorithm for MinSMC achieves approximation ratio $H(\gamma)$, where $\gamma = \max\{f(v) : v \in U\}$ and $H(\gamma) = \sum_{i=1}^{\gamma} 1/i$ is the γ th Harmonic number.*

In this paper, every set C is divided into two disjoint parts: $C = C^1 \cup C^2$, representing vertices of r -value 1 and 2, respectively. Sometimes, we shall use $f(C^1 \cup C^2)$ to emphasize that the calculation of f depends on the division of $C = C^1 \cup C^2$. For two sets $C = C^1 \cup C^2$ and $D = D^1 \cup D^2$, we say that $C \subseteq D$ if $C^1 \subseteq D^1$ and $C^2 \subseteq D^2$. The union of C and D is $C \cup D = (C \cup D)^1 \cup (C \cup D)^2$, where we first set $(C \cup D)^\ell = C^\ell \cup D^\ell$ for $\ell = 1, 2$, and then, for every vertex v with $v \in (C \cup D)^1 \cap (C \cup D)^2$, delete v from $(C \cup D)^1$. For example, if $v \in C^1 \cap D^2$, then v should be put into $(C \cup D)^2$. Such a union can be comprehended as a *priority union*, that is, priority is given to higher r -value. The intersection of C and D is $C \cap D = (C \cap D)^1 \cup (C \cap D)^2$, where $(C \cap D)^\ell = C^\ell \cap D^\ell$ for $\ell = 1, 2$.

2.2 The algorithm

The algorithm for MinRDS is described in Algorithm 1. For a MinRDS instance $G = (V, E)$, we may duplicate every vertex $v \in V$, denoted as v^1 and v^2 ; assign cost $c(v^1) = 1$ and $c(v^2) = 2$. Define $U = V^1 \cup V^2$, $V^1 = \{v^1 : v \in V\}$, $V^2 = \{v^2 : v \in V\}$, and any element subset C can be written as $C = C^1 \cup C^2$ with $C^1 \subseteq V^1$ and $C^2 \subseteq V^2$, where C^1, C^2 are two disjoint sets. A function r associated with C is defined to be

$$r(v) = \begin{cases} 1, & \text{if } v^1 \in C^1, \\ 2, & \text{if } v^2 \in C^2, \\ 0, & \text{if } v^1 \notin C^1 \text{ and } v^2 \notin C^2. \end{cases} \quad (1)$$

A vertex v is said to be *dominated* with respect to C if $v^1 \in C^1$ or $v \in \bigcup_{u^2 \in C^2} N_G[u]$. The potential function f_1 used in Algorithm 1 for MinRDS is defined to be $f_1(C) = |C^1 \cup N[C^2]|$, that is, the number of dominated vertices with respect to C . In line 5 of Algorithm 1, updating r means setting r according to (1), giving priority to higher r -value.

Algorithm 1 Greedy Algorithm for MinRDS

Input: A connected graph G .

Output: A Roman dominating function r for G .

```

1:  $C^1, C^2 \leftarrow \emptyset, C \leftarrow C^1 \cup C^2$ 
2: while  $f_1(C) < n$  do
3:    $v^i \leftarrow \arg \max_{v^i \in U} \frac{\Delta_{v^i} f_1(C)}{i}$ 
4:    $C^i \leftarrow C^i \cup \{v^i\}$ 
5:   Update  $r$  and  $C$ 
6: end while
7: Output  $r$ 
```

The next lemma shows the property of potential function f_1 .

Lemma 2.2 *The function f_1 is normalized, monotone nondecreasing and submodular.*

Proof It is easy to check that $f_1(\emptyset) = 0$, and thus f_1 is normalized. If $C \subseteq D \subseteq U$, then $C^1 \subseteq D^1$ and $N[C^2] \subseteq N[D^2]$. So $f_1(C) \leq f_1(D)$, and thus f_1 is monotone nondecreasing. To prove the submodularity, note that for any two sets $A, B \subseteq V$, we have $N[A \cup B] = N[A] \cup N[B]$ and $N[A \cap B] \subseteq N[A] \cap N[B]$. So, for any two sets $C = C^1 \cup C^2$ and $D = D^1 \cup D^2$, by the definition of f_1 ,

$$\begin{aligned} f_1(C) + f_1(D) &= |C^1 \cup N[C^2]| + |D^1 \cup N[D^2]| \\ &= |(C^1 \cup N[C^2]) \cup (D^1 \cup N[D^2])| + |(C^1 \cup N[C^2]) \cap (D^1 \cup N[D^2])| \\ &= |(C^1 \cup D^1) \cup N[C^2 \cup D^2]| + \\ &\quad + |(C^1 \cap D^1) \cup (C^1 \cap N[D^2]) \cup (N[C^2] \cap D^1) \cup (N[C^2] \cap N[D^2])| \\ &\geq |(C^1 \cup D^1) \cup N[C^2 \cup D^2]| + |(C^1 \cap D^1) \cup N[C^2 \cap D^2]| \\ &= |(C \cup D)^1 \cup N[(C \cup D)^2]| + |(C \cap D)^1 \cup N[(C \cap D)^2]| \\ &= f_1(C \cup D) + f_1(C \cap D). \end{aligned}$$

Hence f_1 is submodular. The lemma is proved. $\square$

Note that line 3 of Algorithm 1 is equivalent to $x \leftarrow \arg \max_{x \in U} \Delta_x f_1(C)/c(x)$. However, Theorem 2.1 is not ready to be used directly. A question is: what if the algorithm chooses both v^1 and v^2 into C , at different iterations? The following lemma shows that this situation will never occur.

Lemma 2.3 *For each vertex $v \in V$, Algorithm 1 will choose at most one copy of v .*

Proof In fact, if v^2 is the first copy of v chosen into C , then afterwards, $\Delta_{v^1} f(C) = 0$ and thus v^1 will no longer be chosen. If v^1 is the first copy of v chosen into C , denote the set C just before v^1 is included as C' , we have $\Delta_{v^1} f(C')/c(v^1) = 1 \geq \Delta_{v^2} f(C')/c(v^2)$, which implies $|N_G[v] \setminus \{u : u^1 \in C'^1\} \cup \bigcup_{u^2 \in C'^2} N_G[u]| = \Delta_{v^2} f(C') \leq 2$, in other words, at most one neighbor of v , say w , is not dominated by C' . Suppose, to the contrary, that v^2 is chosen latter, denote the set C just before v^2 is included as C'' . After v^1 is chosen into C , vertex v is dominated, and thus $\Delta_{v^2} f(C'') \leq 1$. In order that v^2 is able to be chosen, we have $\Delta_{v^2} f(C'') > 0$. Hence $\Delta_{v^2} f(C'') = 1$, which implies that neither w^1 nor w^2 is in C'' . Then, $\Delta_{w^1} f(C'')/c(w^1) \geq 1$. Since $\Delta_{v^2} f(C'')/c(v^2) = 1/2 < \Delta_{w^1} f(C'')/c(w^1)$, v^2 will not be chosen. This contradiction establishes the lemma. $\square$

With the aid of Lemma 2.2 and Lemma 2.3, Algorithm 1 is completely equivalent with the greedy algorithm for MinSMC. By observing that $\gamma = \max\{f_1(x) : x \in U\} = \max\{f_1(v) : v \in V\} \leq \delta_{\max} + 1$, we have the following result from Theorem 2.1.

Theorem 2.4 *Algorithm 1 is an $H(\delta_{\max} + 1)$ -approximation algorithm for MinRDS.*

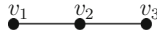


Fig. 1 An example showing non-monotonicity of f_2 . If adding v_2 to C^1 , then $f_2(\{v_2\}) = 4 > 3 = f_2(\emptyset)$. If adding v_2 to C^2 , then $f_2(\{v_2\}) = 2 < 3 = f_2(\emptyset)$.

3 Greedy algorithm for the MinCRDS

Note that a connected graph with the maximum degree $\delta_{\max} \leq 2$ is either a path or a circle. we can easily obtain its minimum connected Roman dominating function. In the following discussion, we consider MinCRDS only for graphs with $\delta_{\max} \geq 3$.

Given a connected graph G with vertex set V , a function r_c associated with $C \subseteq V$ is defined to be

$$r(v) = \begin{cases} 1, & \text{if } v \in C^1, \\ 2, & \text{if } v \in C^2, \\ 0, & \text{if } v \in V \setminus (C^1 \cup C^2). \end{cases} \quad (2)$$

We define the potential function f_2 as follows. For a vertex set $C \subseteq V$, denote by $p(C)$ the number of components of induced subgraph $G|_C$. Let $E_N(C)$ be the set of edges with one end in $N_G(C^2) \setminus C$ and another end in C . We use $q(C)$ to denote the number of components of the graph $G|_{(C)}$ on vertex set V and edge set $E(G|_C) \cup E_N(C)$. Define $f_2(C) = p(C) + q(C)$. The function f_2 has the following property.

Lemma 3.1 *A vertex set $C \subseteq V$ is a CRDS of G if and only if $f_2(C) = 2$.*

Proof Note that for any vertex set C , we have $p(C), q(C) \geq 1$. So, $f_2(C) = 2$ if and only if $p(C) = q(C) = 1$. If C is a CRDS, then $G|_C$ is connected (hence $p(C) = 1$), and every vertex $v \in V \setminus C$ is adjacent with a vertex in $C^2 \subseteq V(G|_C)$, and thus $G|_{(C)}$ is connected (so $q(C) = 1$). Conversely, if $p(C) = q(C) = 1$, then $G|_C$ is connected. If there is a vertex $v \in V \setminus C$ which is not adjacent with any vertex of C^2 , then v is an isolated vertex in $G|_{(C)}$, and thus $q(C) \geq 2$, a contradiction. So, C is a CRDS. $\square$

Remark 3.2 Note that f_2 is neither monotone non-increasing nor monotone non-decreasing, as shown by Fig. 1.

One natural idea is to use the technique in the previous section: treat every vertex v as two copies v^1, v^2 with costs 1 and 2, respectively. But one difficulty occurs: unlike Lemma 2.3, after v^1 is chosen, it is still possible to choose v^2 . So, we have to think out another greedy algorithm to circumvent this difficulty.

Our greedy algorithm for MinCRDS is described in Algorithm 2. We allow each vertex in V to be selected multiple times, but *at most twice*, in the algorithm. In line 3 of Algorithm 2, the multiset X may contain ℓ copies of vertex v , where $\ell \in \{0, 1, 2\}$, and the contribution of vertex v to $|X|$ is ℓ . For instance, if the algorithm chooses $X = \{v_1, v_1, v_3\}$ in some iteration, then $|X| = 3$. Note that every vertex can be selected at most twice. So, if v is already selected twice, then subsequent X can no longer select v ; if v is already selected once, then v may be selected at most one another time by subsequent X . For a set $C = C^1 \cup C^2$ and a multiset X , define the

new set (not multiset)

$$(C + X) = (C + X)^1 \cup (C + X)^2,$$

where $(C + X)^1 = \{v \mid v \in C^1 \text{ and } v \notin X\} \cup \{v \mid v \notin C \text{ and } v \text{ occurs once in } X\}$, $(C + X)^2 = \{v \mid v \in C \text{ or } v \in X\} \setminus (C + X)^1$. That is, if $v \notin C$, then add v into C^ℓ if v is selected ℓ times by X ; if v is already in C^1 , and is selected another time by X , then move v into C^2 ; note that if $v \in C^2$, then v cannot be selected by X . For two subsets $A, C \subseteq V$, define the *variant marginal profit* of A over C as

$$\Delta_{+A} f_2(C) = f_2(C + A) - f_2(C).$$

The algorithm has a parameter k related with the error term ε which will be determined later. Line 3 of Algorithm 2 chooses a multiset X with $|X| \leq 2k + 1$ that maximizes $\frac{f_2(C) - f_2(C+X)}{|X|}$. The update of C^1 and C^2 is in the way of defining $C + X$. The update of r_c is according to (2).

Algorithm 2 Greedy Algorithm for MinCRDS

Input: A connected graph G and a positive integer $k \geq 2$.

Output: A connected Roman dominating function r_c for G .

```

1:  $C^1, C^2 \leftarrow \emptyset, C = C^1 \cup C^2$ 
2: while  $f_2(C) > 2$  do
3:    $X \leftarrow \arg \max\{X: |X| \leq 2k-1\} - \frac{\Delta_{+X} f_2(C)}{|X|}$ 
4:    $C \leftarrow C + X$ 
5:   Update  $C^1, C^2$  and  $r_c$ 
6: end while
7: Output  $r_c$ 
```

The following lemma shows that before the f_2 -value drops to 2, there always exists a vertex the addition of which, either into C^1 or C^2 , strictly reduces the f_2 -value.

Lemma 3.3 Suppose $G = (V, E)$ is a connected graph on at least three vertices, $C = C^1 \cup C^2$ is a vertex set with $f_2(C) > 2$. Then there exists a vertex $v \in V \setminus C$ such that $f_2(C + \{v\}) < f_2(C)$ or $f_2(C + \{v, v\}) < f_2(C)$.

Proof By Lemma 3.1, C is not a CRDS. There are two cases to be considered: C is not an RDS, and C is an RDS but is not connected.

Case 1 C is not an RDS.

In this case, $C \cup N_G(C^2) \neq V$. By the connectedness of G , there is a vertex $v \in V \setminus (C \cup N_G(C^2))$ which is adjacent with a vertex $u \in C \cup N_G(C^2)$.

If v is adjacent with a vertex of $u \in C$, then u must belong to C^1 (otherwise v will belong to $N_G(C^2)$, contradicting our assumption on v). Adding v into C^1 , we have $p(C + \{v\}) \leq p(C)$. Furthermore, since v is an isolated vertex in $G|_{\{C\}}$, and in $G|_{\{C+\{v\}\}}$, v is merged into the connected component of $G|_{\{C\}}$ containing u , so $q(C + \{v\}) < q(C)$. It follows that $f_2(C + \{v\}) < f_2(C)$.

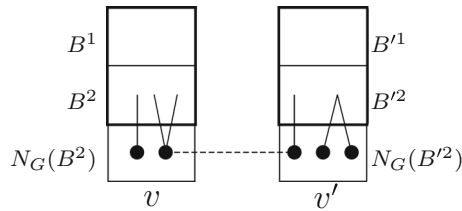


Fig. 2 An illustration for the proof of Case 2 when no connected components of $G|_C$ have a common neighbor. A blackened square indicates a connected component of $G|_C$. With those vertices in $N_G(B^2)$ attached, we have a connected component of $G|_C$

If v is not adjacent with any vertex in C , then $u \in N_G(C^2)$. Adding u into C^2 , we have $p(C + \{u, u\}) \leq p(C)$. Furthermore, uv becomes an edge in $E_N(C + \{u, u\})$, connecting the isolated vertex v in $G|_C$ to the connected component of $G|_C$ containing u . Hence $q(C + \{u, u\}) < q(C)$, and thus $f_2(C + \{u, u\}) < f_2(C)$.

Case 2 C is an RDS, but $G|_C$ is not connected.

If there are two connected components of $G|_C$ which have a common neighbor v , then adding v into C^1 merges all those connected components of $G|_C$ adjacent with v , and thus $p(C + \{v\}) < p(C)$. While $q(C + \{v\}) \leq q(C)$.

If no connected components of $G|_C$ have a common neighbor, then $G|_C$ is not connected, and every connected component of $G|_C$ has the form $H_B = G|_B + N_G(B^2)$, the meaning of which is that H_B is composed of a connected component of $G|_C$, which is induced by vertex set B , with vertices in $N_G(B^2)$ attached to $G|_B$ (see Fig. 2). Furthermore, the nearest components of $G|_C$, say H_B and $H_{B'}$, are linked by an edge vv' with $v \in N_G(B^2)$ and $v' \in N_G(B'^2)$. Adding v into C^1 , vv' becomes an edge in $E_N(C + \{v\})$, which can merge H_B and $H_{B'}$ into a larger connected subgraph of $G|_{C+\{v\}}$. So, $q(C + \{v\}) < q(C)$, while $p(C + \{v\}) = p(C)$. In any case, we have $f_2(C + \{v\}) < f_2(C)$. $\square$

Remark 3.4 There exist cases in which adding any vertex v into C^1 cannot reduce the f_2 -value. Consider Fig. 1 again. Note that $f_2(\emptyset) = 3$, while adding any vertex v into C^1 , we have $f_2(\emptyset + \{v\}) = 4$. So, in order that the f_2 -value of Algorithm 2 can continuously decrease, in other words, in order to continuously increase the feasibility of C , selecting more than one vertex into multiset X is necessary.

Suppose Algorithm 2 adds multisets $X_1, X_2, \dots, X_g$ to C in sequence. Denote $C_i = X_1 + X_2 + \dots + X_i$ for $i = 1, \dots, g$, and let $C_0 = \emptyset$. Lemma 3.3 directly leads to the following corollary.

Corollary 3.5 For $i = 0, 1, \dots, g - 1$,

$$\frac{-\Delta_{+X_{i+1}} f_2(C_i)}{|X_{i+1}|} \geq \frac{1}{2}.$$

Proof Since f_2 is an integer-valued function, $f_2(C + \{v\}) < f_2(C)$ or $f_2(C + \{v, v\}) < f_2(C)$ implies $-\Delta_{+X} f_2(C) \geq 1$ for $X = \{v\}$ or $X = \{v, v\}$, leading to cost-effectiveness $-\frac{\Delta_{+X} f_2(C)}{|X|} \geq \frac{1}{2}$. Then the lemma follows from the greedy strategy. $\square$

For the connected graph $G = (V, E)$, suppose r_c^* is the minimum connected Roman dominating function of G . Denote $C^* = \{v \in V \mid r_c^*(v) = 1 \text{ or } r_c^*(v) = 2\}$. The following lemma shows a decomposition of C^* (any pair of parts may have at most one common vertex), which established a bridge to compare C^* and the output CRDS of Algorithm 2.

Lemma 3.6 *For any integer $k \geq 2$, C^* can be decomposed into $C^* = Y_1 \cup Y_2 \cup \dots \cup Y_h$ for some $h \geq 1$, satisfying the following three conditions:*

- (i) $G|_{Y_j}$ and $G|_{Y_1 \cup Y_2 \cup \dots \cup Y_j}$ are both connected for $j = 1, \dots, h$;
- (ii) $1 \leq r_c^*(Y_j) \leq 2k - 1$ for $j = 1, \dots, h$, and at least $h - 1$ parts satisfy $r_c^*(Y_j) \geq k + 1$;
- (iii) $r_c^*(Y_1) + r_c^*(Y_2) + \dots + r_c^*(Y_h) \leq r_c^*(C^*) + 2(h - 1)$.

Proof Let T be a spanning tree of $G|_{C^*}$ (such T exists because $G|_{C^*}$ is connected), and choose an arbitrary vertex $v \in C^*$ as the root of T . For any vertex $x \in C^*$, denote by $T(x)$ the subtree of T rooted at x , and $r_c^*(T(x)) = \sum_{u \in V(T(x))} r_c^*(u)$.

If $r_c^*(T) \leq 2k - 1$, then let $h = 1$ and $Y_1 = C^*$. The three conditions are satisfied.

If $r_c^*(T) \geq 2k$, then there must exist a vertex x with $r_c^*(T(x)) \geq k + 1$ such that all of its children, say $y_1, \dots, y_t$, satisfy $r_c^*(T(y_i)) \leq k$. We shall iteratively construct a connected vertex set Y and then continue to consider the decomposition of tree T' , with only one vertex common to Y and T' .

First, suppose $r_c^*(x) = 1$. If one child y_i has $r_c^*(T(y_i)) = k$, then let $Y = T(y_i) \cup \{x\}$ and $T' = T - T(y_i)$. Clearly, $r_c^*(Y) = k + 1$. If $r_c^*(T(y_i)) \leq k - 1$ ($i = 1, \dots, t$), since $r_c^*(x) = 1$ and $r_c^*(T(x)) \geq k + 1$ implies $\sum_{i=1}^t r_c^*(T(y_i)) \geq k$, there is an index j satisfying

$$r_c^*(T(y_1)) + \dots + r_c^*(T(y_j)) \leq k - 1 \quad (3)$$

and

$$r_c^*(T(y_1)) + \dots + r_c^*(T(y_{j+1})) \geq k. \quad (4)$$

Let $Y = T(y_1) \cup \dots \cup T(y_{j+1}) \cup \{x\}$ and $T' = T - (T(y_1) \cup \dots \cup T(y_{j+1}))$. Note that $r_c^*(Y) \geq k + 1$ because of (4), and $r_c^*(Y) \leq 2k - 1$ because of $r_c^*(T(y_{j+1})) \leq k - 1$ and (3).

Next, suppose $r_c^*(x) = 2$. If one child y_i has $r_c^*(T(y_i)) = k - 1$, then let $Y = T(y_i) \cup \{x\}$ and $T' = T - T(y_i)$. If $r_c^*(T(y_i)) \leq k - 2$ for any $i = 1, \dots, t$. Similar to the above analysis, there exists an index j with $r_c^*(T(y_1)) + \dots + r_c^*(T(y_j)) \leq k - 2$ and $r_c^*(T(y_1)) + \dots + r_c^*(T(y_{j+1})) \geq k - 1$. Let $Y = T(y_1) \cup \dots \cup T(y_{j+1}) \cup \{x\}$ and $T' = T - (T(y_1) \cup \dots \cup T(y_{j+1}))$.

Iteratively using such a construction method, we obtain a sequence of sets Y . If ordered in the reverse order of construction, then they satisfy conditions (i) and (ii). Note that in the above construction, x is the only vertex common to Y and T' . So, every time a Y is construction (except for the last Y when the remaining tree T'

has $r_c^*(T') \leq 2k - 1$, Y contains one vertex which will appear again in the latter decomposition. Combining this with $r_c^*(x) \leq 2$, condition (iii) is satisfied. $\square$

For a decomposition of C^* as in Lemma 3.6, denote $C_j^* = Y_1 \cup \dots \cup Y_j$ for $1 \leq j \leq h$, and let $C_0^* = \emptyset$. We have the following property.

Lemma 3.7 For any $0 \leq i \leq g - 1$ and $2 \leq j \leq h$,

$$-\Delta_{Y_j} f_2(C_i \cup C_{j-1}^*) \leq -\Delta_{+Y_j} f_2(C_i) + 2.$$

Proof Note that $-\Delta_{Y_j} f_2(C_i \cup C_{j-1}^*) = f_2(C_i \cup C_{j-1}^*) - f_2(C_i \cup C_{j-1}^* \cup Y_j) = f_2(C_i \cup C_{j-1}^*) - f_2(C_i \cup C_j^*)$ and $-\Delta_{+Y_j} f_2(C_i) = f_2(C_i) - f_2(C_i + Y_j)$. We first prove two claims.

Claim 1 $-\Delta_{Y_j} p(C_i \cup C_{j-1}^*) \leq -\Delta_{+Y_j} p(C_i) + 1$.

Note that for any vertex set C and a connected vertex set Y , both $-\Delta_Y p(C)$ and $-\Delta_{+Y} p(C)$ equal the number of connected components of $G|_C$ that are adjacent with Y minus 1 (if a connected component of $G|_C$ has a common vertex with Y , then it is also regarded as adjacent with Y). By (i) of Lemma 3.6, $G|_{Y_j}$ is connected, and C_{j-1}^* induces a connected subgraph. So, compared with $G|_{C_i}$, subgraph $G|_{C_i \cup C_{j-1}^*}$ has at most one more connected component which is adjacent with Y_j , namely the one that contains C_{j-1}^* . Then, Claim 1 follows.

Claim 2 $-\Delta_{Y_j} q(C_i \cup C_{j-1}^*) \leq -\Delta_{+Y_j} q(C_i) + 1$.

For a vertex set C and a connected vertex set Y , we first consider the effect of $-\Delta_Y q(C)$ and $-\Delta_{+Y} q(C)$.

In the case $Y \cap C = \emptyset$, the effect of $-\Delta_Y q(C)$ and $-\Delta_{+Y} q(C)$ are the same. A connected component of $G|_{\langle C \rangle}$ with the form $H_B = G|_B + N_G(B^2)$ (this notation is defined in the proof for Case 2 of Lemma 3.3) can be merged into $H_Y = G|_Y + N_G(Y^2)$ if and only if either B and Y are adjacent, or $N_G(B^2)$ and Y are adjacent, or B and $N_G(Y^2)$ are adjacent (having a common vertex is viewed as adjacent). A trivial component of $G|_{\langle C \rangle}$, which is a vertex v not dominated by C , can be merged into H_Y if and only if $v \in Y \cup N_G(Y^2)$. Call the above component H_B or the above trivial component v as an effective component of $G|_{\langle C \rangle}$ with respect to Y . Then, both $-\Delta_Y q(C)$ and $-\Delta_{+Y} q(C)$ equal to the number of effective components with respect to Y .

In the case when $Y \cap C \neq \emptyset$, one big difference occurs when a vertex $v \in Y \cap C$ has its r_c -value 1 in both C and Y . For $-\Delta_Y f_2(C)$, the contribution of $\cup\{v\}$ is the same as the above argument. While for $-\Delta_{+Y} f_2(C)$, v is moved from value 1 in C^1 to value 2 in $(C + Y)^2$. So v can dominate more vertices, and the consequence of $+\{v\}$ might merge more effective components than $\cup\{v\}$. For the case $r_c(v) = 1$ in C and $r_c(v) = 2$ in Y , both the effect of $\cup\{v\}$ and $+\{v\}$ is to move v from value 1 to value 2. And for the case when $r_c(v) = 2$ in C , we may ignore the effect of $\cup\{v\}$ and $+\{v\}$.

Recall that both $G|_{Y_j}$ and $G|_{C_{j-1}^*}$ are connected. So, $C_i \cup C_{j-1}^*$ may have at most one more effective component with respect to Y_j than C_i , namely the connected component containing C_{j-1}^* . From the above argument, Claim 2 follows.

Then the lemma follows from combining the above two claims. $\square$

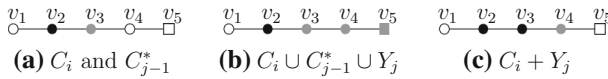


Fig. 3 An example showing that Lemma 3.7 is tight. In (a), $C_i^2 = \{v_2\}$ is marked by a blackened vertex, $C_i^1 = \{v_3\}$ is marked by a grey vertex, $C_{j-1}^* = \{v_5\}$ is marked by a square, and suppose $r_c(v_5) = 1$. It can be calculated that $p(C_i) = 1$, $q(C_i) = 3$, $p(C_i \cup C_{j-1}^*) = 2$, $q(C_i \cup C_{j-1}^*) = 3$. Suppose $Y_j = \{v_3, v_4\}$ and every vertex in Y_j has r_c -value 1. By (b) which depicts $C_i \cup C_{j-1}^* \cup Y_j$, we have $p(C_i \cup C_{j-1}^* \cup Y_j) = q(C_i \cup C_{j-1}^* \cup Y_j) = 1$. So, $-\Delta_{Y_j} f_2(C_i \cup C_{j-1}^*) = 3$. In (c) which depicts $C_i + Y_j$, v_3 becomes a blackened vertex, having r_c -value 2 in $C_i + Y_j$. It can be calculated that $p(C_i + Y_j) = 1$ and $q(C_i + Y_j) = 2$, and thus $-\Delta_{+Y_j} f_2(C_i) = 1$.

Remark 3.8 The inequality in Lemma 3.7 is tight, as can be shown by Fig. 3.

The following lemma gives a relationship between the sizes of V and C^* .

Lemma 3.9 $|V| \leq \delta_{\max}|C^*| + 1$.

Proof The lemma follows from the fact that any *connected* subset $C \subseteq V$ can dominate at most $\delta_{\max}|C| + 1$ vertices. We prove the fact by induction on $|C|$.

For $|C| = 1$, suppose v is the only vertex in C . If $v \in C^1$, then v only dominates one vertex, namely v itself. If $v \in C^2$, then v dominates $N_G[v]$, which contains $\deg_G(v) + 1 \leq \delta_{\max} + 1$ vertices.

For $|C| \geq 2$, let v be a vertex of C such that $G|_{C \setminus \{v\}}$ is connected (such v exists since C is a connected vertex set). Suppose $C \setminus \{v\}$ dominates at most $\delta_{\max}(|C| - 1) + 1$ vertices. Adding v into $(C \setminus \{v\})^1$ increases the number of dominated vertices by at most 1. Adding v into $(C \setminus \{v\})^2$ increases the number of dominated vertices by at most $(\deg_G(v) + 1) - \deg_{C \setminus \{v\}}(v) \leq \delta_{\max}$ (note that v is adjacent with some vertex in $C \setminus \{v\}$ and thus $\deg_{C \setminus \{v\}}(v) \geq 1$). It follows that C dominates at most $\delta_{\max}(|C| - 1) + 1 + \delta_{\max} = \delta_{\max}|C| + 1$ vertices. The lemma is proved. $\square$

Now, we are ready to prove the approximation ratio of Algorithm 2.

Theorem 3.10 For any real number $\varepsilon > 0$, setting $k = \lceil \frac{2}{\varepsilon} + 1 \rceil$ in Algorithm 2, the approximation ratio for MinCRDS is at most $(1 + \varepsilon)(2 + \ln \delta_{\max})$, where $\delta_{\max}$ is the maximum degree of the input graph.

Proof Suppose Algorithm 2 selects vertex sets $X_1, \dots, X_g$ in sequence. Let $C_i = X_1 + \dots + X_i$ for $i = 1, \dots, g$ and let $C_0 = \emptyset$. Note that C_g is the output of the algorithm. Let r_c^* be a minimum weight CRDS function. Denote $opt = \sum_{v \in V} r_c^*(v)$ the optimal value. For $i = 0, 1, 2$, let $V_{r_c^*}^i = \{v \in V \mid r_c^*(v) = i\}$ and $C^* = V_{r_c^*}^1 \cup V_{r_c^*}^2$. Decompose C^* into $Y_1 \cup \dots \cup Y_h$ as in Lemma 3.6. Let $C_j^* = Y_1 \cup \dots \cup Y_j$ for each $j = 1, \dots, h$ and $C_0^* = \emptyset$.

For the choice of X in line 3 of Algorithm 2, recall that $|X|$ refers to the cardinality of multi-set X . We may regard each $r_c^*(Y_j)$ as $|Y_j|$ if Y_j is viewed as a multi-set, that is, each $v \in Y_j$ has $r_c^*(v)$ copies in Y_j . Then $r_c^*(Y_j) \leq 2k - 1$ (condition (ii) of Lemma 3.6) implies that Y_j is a valid candidate for X . By the greedy choice of X , we have

$$\frac{-\Delta_{+X_{i+1}} f_2(C_i)}{|X_{i+1}|} \geq \frac{-\Delta_{+Y_j} f_2(C_i)}{|Y_j|} \quad (\forall 0 \leq i \leq g-1, 1 \leq j \leq h). \quad (5)$$

Combining this with Lemma 3.7, condition (iii) of Lemma 3.6, and the observation $-\Delta_{Y_j} f_2(C_i \cup C_{j-1}^*) \leq -\Delta_{Y_j} f_2(C_i)$ for $j = 1$, we have

$$\begin{aligned} \frac{-\Delta_{X_{i+1}} f_2(C_i)}{|X_{i+1}|} &\geq \frac{-\sum_{j=1}^h \Delta_{Y_j} f_2(C_i)}{\sum_{j=1}^h |Y_j|} \\ &\geq \frac{-2(h-1) - \sum_{j=1}^h \Delta_{Y_j} f_2(C_i \cup C_{j-1}^*)}{\sum_{j=1}^h |Y_j|} \\ &\geq \frac{-2(h-1) - (f_2(C_i \cup C_h^*) - f_2(C_i))}{opt + 2(h-1)} \\ &= \frac{f_2(C_i) - 2h}{opt + 2(h-1)}, \end{aligned} \quad (6)$$

where the last equality uses the observation that $C_i \cup C_h^* \supseteq C^*$ is also a CRDS of G , and thus $f_2(C_i \cup C_h^*) = 2$ by Lemma 3.1.

Denote $a_i = f_2(C_i) - 2h$. Note that a_i is monotone decreasing with respect to i , and $a_0 = |V| - 2h$. Inequality (6) leads to

$$\frac{a_i - a_{i+1}}{|X_{i+1}|} \geq \frac{a_i}{opt + 2(h-1)}, \quad (7)$$

and thus

$$a_{i+1} \leq a_i \left(1 - \frac{|X_{i+1}|}{opt + 2(h-1)} \right) \leq a_i e^{-\frac{|X_{i+1}|}{opt + 2(h-1)}}, \quad (8)$$

where the second inequality uses the fact $1 + x \leq e^x$ for any real number x . Recursively using inequality (8), we have

$$a_{i+1} \leq a_0 e^{-\frac{(|X_{i+1}| + |X_i| + \dots + |X_1|)}{opt + 2(h-1)}} \quad (9)$$

Case 1 $|V| \geq opt + 2h$.

In this case, $a_0 \geq opt$ and $a_g = 2 - 2h < opt$. Hence there exists an index $i_0 \in \{0, 1, \dots, g-1\}$ such that

$$a_{i_0} \geq opt > a_{i_0+1}.$$

Let $b = a_{i_0} - opt$ and $b' = opt - a_{i_0+1}$. Write $|X_{i_0+1}| = d + d'$ such that

$$\frac{b}{d} = \frac{b'}{d'} = \frac{a_{i_0} - a_{i_0+1}}{|X_{i_0+1}|}. \quad (10)$$

Note that for the computed CRDS function r_c , its value $\sum_{v \in V} r_c(v) = |C_g|$ can be divided into two parts: $\sum_{i=1}^{i_0} |X_i| + d$ and $d' + \sum_{i=i_0+2}^g |X_i|$.

First, we prove

$$\sum_{i=1}^{i_0} |X_i| + d \leq (opt + 2(h-1)) \ln \frac{|V| - 2h}{opt}. \quad (11)$$

Combining (7) and (10), we have

$$\frac{b}{d} = \frac{a_{i_0} - opt}{d} \geq \frac{a_{i_0}}{opt + 2(h-1)},$$

resulting in

$$opt \leq a_{i_0} \left(1 - \frac{d}{opt + 2(h-1)} \right) \leq a_{i_0} e^{-\frac{d}{opt + 2(h-1)}}.$$

Combining this with (9),

$$opt \leq a_0 e^{-\frac{d + |X_{i_0}| + \dots + |X_1|}{opt + 2(h-1)}} = (|V| - 2h) e^{-\frac{d + |X_{i_0}| + \dots + |X_1|}{opt + 2(h-1)}}.$$

By a rearrangement, inequality (11) follows.

Next, we show

$$d' + \sum_{i=i_0+2}^g |X_i| \leq 2(opt + 2(h-1)). \quad (12)$$

By Lemma 3.5, $\frac{-\Delta_{+X_{i+1}} f_2(C_i)}{|X_{i+1}|} \geq \frac{1}{2}$, which leads to $|X_{i+1}| \leq -2\Delta_{+X_{i+1}} f_2(C_i)$. By (10) and the observation $\frac{a_{i_0} - a_{i_0+1}}{|X_{i_0+1}|} = \frac{-\Delta_{+X_{i_0+1}} f_2(C_{i_0})}{|X_{i_0+1}|} \geq \frac{1}{2}$, we have $d' \leq 2b'$. So,

$$\begin{aligned} d' + \sum_{i=i_0+2}^g |X_i| &\leq 2 \left(b' - \sum_{i=i_0+2}^g \Delta_{+X_i} f_2(C_{i-1}) \right) \\ &= 2(b' + f_2(C_{i_0+1}) - f_2(C_g)) \\ &= 2(opt - a_{i_0+1} + f_2(C_{i_0+1}) - 2) \\ &= 2(opt + 2(h-1)). \end{aligned}$$

Inequality (12) is proved.

Combining inequalities (11) and (12), we have

$$\sum_{v \in V} r_c(v) = \sum_{i=1}^g |X_i| \leq (opt + 2(h-1)) \left(2 + \ln \frac{|V| - 2h}{opt} \right). \quad (13)$$

By Lemma 3.6,

$$(h-1)(k+1) + 1 \leq r_c^*(Y_1) + \dots + r_c^*(Y_h) \leq opt + 2(h-1).$$

It follows that

$$h - 1 \leq \frac{opt}{k - 1}. \quad (14)$$

By Lemma 3.9, $|V| \leq \delta_{\max} opt + 1$, and thus

$$\frac{|V| - 2h}{opt} \leq \delta_{\max} + \frac{1 - 2h}{opt} \leq \delta_{\max} \quad (15)$$

(note that $h \geq 1$). Substituting (14) and (15) into (13), we have

$$\sum_{v \in V} r_c(v) \leq opt \left(1 + \frac{2}{k - 1}\right) (2 + \ln \delta_{\max}) \leq opt(1 + \varepsilon)(2 + \ln \delta_{\max}).$$

Case 2 $|V| < opt + 2h$.

Using Lemma 3.5, and similar to the deduction of (12), we have

$$\sum_{v \in V} r_c(v) = \sum_{i=1}^g |X_i| \leq 2(f_2(C_0) - f_2(C_g)) = 2(|V| - 2) < 2(opt + 2(h - 1)).$$

Then by (14) and the choice of k ,

$$\sum_{v \in V} r_c(v) < 2(1 + \varepsilon)opt.$$

The desired approximation ratio is proved. $\square$

4 Conclusions

In this paper, we use a greedy strategy to design an $H(\delta_{\max} + 1)$ -approximation algorithm for MinRDS, and a $(1 + \varepsilon)(2 + \ln \delta_{\max})$ -approximation algorithm for MinCRDS. The former improves previous approximation ratio $2 + 2 \ln(\delta_{\max} + 1)$ in [11]. The challenge for the latter is how to deal with a non-monotone non-submodular potential function. Note that $(1 + \varepsilon)(2 + \ln \delta_{\max}) = (1 + \varepsilon) \ln \delta_{\max} + O(1)$. A question is: does MinCRDS admit an approximation ratio of $\ln \delta_{\max} + O(1)$?

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